

## \$20 Element C Worksheet

### Design Requirement 1 (High Priority)

- The solution must fit the client making the cart easier and more comfortable to use.
- The solution must make the cart more comfortably usable for the client who is 6'3".  
When we spoke with the client they stated that the problem they faced was having to bend over to push the cart which caused lower back pain. The solution must make the cart easier for the client to use with less to no back pain.

### Design Requirement 2 (High Priority)

- The solution has zero permanent impact on the cart.
- One requirement we had was that the cart couldn't be impacted by the solution so that the cart can be returned back to its original state at time. This allows for one unit to be used as a solution for multiple of the same carts. The client stated that they would prefer if it is quickly and easily removable without bending over.

### Design Requirement 3 (High Priority)

- The solution must not slow down the user as they push the cart around with materials.
- We found that the user would be less likely to use the solution if it slows down the use of the cart wasting their time. The solution is useless if the intended user doesn't want to use the solution to help with the cart. When we talked with the client they stated that they tend to walk quickly even with a loaded cart and would not be impressed with a design that would slow down the use of the cart.

### Design Requirement 4 (High Priority)

- The solution must be able to withstand pushing the max load of the cart (500 lbs) up a sloped ramp.
- The solution is not useful if it limits the usefulness of the cart including limiting the max weight that the cart can be loaded to. The cart needs to be used to its full potential as it can be loaded with heavy objects such as heavy textbooks. The client stated that the max load of the carts is not commonly used but is occasionally needed so the solution must be able to handle the force without issue.

### Design Requirement 5 (High Priority)

- The solution must not reduce the usable area of the cart.
- The end user will not want to use the solution often if at all if it reduces the area of the cart they can use. These carts are used to move large objects and the maximum area of these carts are used more than the max load of the carts so the area not being affected is a very important point. The client stated that they would not find a solution useful if it took up space on the cart. The cart would be less useful to move items around.

#### Design Requirement 6 (Low Priority)

- The solution must appear well made and not stand out from the cart.
- We found that end users don't want a solution that sticks out and looks bad in its environment. People prefer to use items that are visually appealing rather than looking as if looks were an afterthought if at all. When we talked with the client they stated that the solution needs to look completed and not stick out so that it is more desirable to look at and use.

#### Design Requirement 7 (Low Priority)

- The solution must last greater than one year in normal use.
- End users do not want to get a solution for a problem if it doesn't have longevity in its design. An end user who spends money on a solution doesn't want one that will only last a few weeks or months before breaking and leaving them in the same place as they were before. As we talked with the client they stated that they need it to last at least a year and ideally between four and five years. This made us consider the longevity of the solution for the design.

#### Design Requirement 8 (Low Priority)

- The design must dampen vibrations and small bumps as the cart is pushed.
- The end solution should be comfortable to use so the users hands don't cramp up. If the solution is too stiff the user's hands will tire out and cramp up thus making the customer not want to use the product. When we spoke with the client they stated that they wanted enough flex to remove vibrations but not so much that the handle feels weak or unsecure.

#### Design Requirement 9 (Low Priority)

- The design must be adjustable for multiple carts and users
- The solution should be usable for multiple carts and people. Adaptability for the solution will make it a better overall solution and useful for more people and carts. A more universal solution will make the product more desirable for the end consumer with confidence that it will work for their application. We spoke with the client over this point and they stated that they had multiple carts with different handle designs and a universal solution for any of the carts would be much more beneficial.